Near-term Quantum Computing — lecture notes

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Table of contents

Near-term Quantum Computing 3
How to use these notes . . . 3
Prerequisites . . . 3
Tutorials . . . 4

1 Platforms, gates and measurement 5
1.1 Noisy qubits and fault tolerance . . . 5
1.2 Physical qubits . . . 7
1.2.1 Requirements for a quantum computer . . . 7
1.2.2 Energy levels, frequencies and pulses . . . 8
1.2.3 Superconducting qubits . . . 8
1.2.4 Trapped ions . . . 12
1.2.5 Other platforms . . . 12
1.3 Native gates and connectivity . . . 14
1.3.1 Pauli rotations . . . 15
1.3.2 Physical implementation of gates . . . 16
1.3.3 Native gate sets . . . 18
1.3.4 Compiling into native gates . . . 19
1.3.5 Virtual Z gates . . . 21
1.3.6 Connectivity . . . 23
1.4 Sources of noise . . . 24
1.5 Measurement . . . 26
1.5.1 Readout . . . 26
1.5.2 Shots and expectation values . . . 26
1.5.3 Shot noise . . . 27
1.5.4 Timescales . . . 27
1.6 More exercises . . . 28
1.7 Further reading . . . 30
References . . . 30

Appendices 32

A Glossary 32

Near-term Quantum Computing

Lecture notes for *Near-term Quantum Computing* (UvA, 2026-2027), one chapter per lecture. Chapters are added as the course goes on.

How to use these notes

- **Chapters.** Each chapter follows one lecture and goes into more detail than the lecture can.
- **Glossary.** Underlined terms are explained in the Glossary. On the website, hover over a term (or tap it on a phone) to see its definition, and click it to open the glossary.
- **Search.** The search box at the top of the sidebar searches all chapters; on a computer, the shortcut is `/`.
- **PDF.** The download icon next to the title gives all published chapters as one PDF, with the same numbering, for printing or reading offline. Glossary terms are in italics there, and the glossary is an appendix.

Prerequisites

The notes assume basic quantum mechanics and the circuit model: Dirac notation, Pauli matrices, density matrices, expectation values and their statistical error.

The tutorials are Jupyter notebooks in Python, which run in Google Colab. They assume basic Python (variables, functions, loops, lists and dictionaries) and a little familiarity with Jupyter notebooks, NumPy arrays and Matplotlib plots. If any of these is new to you, these short introductions cover what we need:

- Python: the official Python tutorial, sections 3 to 5;
- Jupyter notebooks in Colab: Overview of Colab;
- NumPy: NumPy: the absolute basics for beginners;
- Matplotlib: Quick start guide.

Tutorials

Each notebook opens in Google Colab. To keep your work, use **File → Save a copy in Drive** before you start. If you prefer to work in your own environment, download the notebook instead. New notebooks and solutions appear in this table as they are released.

Tutorial	Date	Topic	Notebook	Solutions
T1	Fri 30 Oct	Circuits, simulators and sampling	Colab · Download	*after the tutorial*
T2	Fri 6 Nov		*not yet released*	*after the tutorial*
T3	Fri 13 Nov		*not yet released*	*after the tutorial*
T4	Fri 20 Nov		*not yet released*	*after the tutorial*
T5	Fri 27 Nov		*not yet released*	*after the tutorial*
T6	Fri 4 Dec		*not yet released*	*after the tutorial*
T7	Fri 11 Dec		*not yet released*	*after the tutorial*

Version: 2026-10-16 (draft).

1 Platforms, gates and measurement

On paper, a quantum algorithm is a circuit: gates on ideal qubits, followed by measurements. A real quantum computer is a physical device, whose physics decides which operations it can perform, between which qubits, how fast and how reliably. In this chapter, we look at this physical side and at what it means for the circuits we can run.

By the end of the chapter, you should be able to explain what a quantum computer needs from its qubits and how superconducting circuits and trapped ions provide it; why devices have different native gates, why Z rotations are usually free, and what limited connectivity costs; and how many shots an expectation value needs and how long they take.

1.1 Noisy qubits and fault tolerance

Every physical qubit is imperfect. Each gate has a small chance of going wrong, the stored quantum state degrades even when nothing is done to it, and measurements sometimes return the wrong result. These errors add up. If each gate fails with probability 0.1%, a circuit of a thousand gates runs without any error only about 37% of the time, since $0.999^{1000} \approx e^{-1}$. Noise therefore limits the size of the circuits that a device can run usefully.

Preskill called the devices of this era *NISQ* devices, for *noisy intermediate-scale quantum* (Preskill 2018). *Intermediate scale* means tens to thousands of qubits, enough that many of their circuits are hard to simulate on a classical computer, but far from the millions of qubits that the best-known quantum algorithms need. *Noisy* means that their errors are not corrected.

Getting useful results from a NISQ device means working within the limit set by noise. Circuits are kept short, so that errors have little chance to build up. The quantities to keep small are the circuit's *depth*, the number of layers of gates that must run one after another, and its number of two-qubit gates, the least reliable operations on current devices (Section 1.4). Each run of a circuit returns a single random outcome, so every quantity is estimated from many repetitions, called shots (Section 1.5.2). And *error mitigation* reduces the effect of the remaining noise on these estimates, without correcting the errors themselves, by running extra or modified circuits and combining their results. Error mitigation is not the same as error correction: it only works if the circuit does not accumulate too much error in the first place, and its price is running more shots. We come back to it later in the course.

Correcting errors is possible in principle. *Quantum error correction* stores each qubit of information redundantly in many *physical qubits*, which together form one *logical qubit*. Repeated checks on the physical qubits reveal errors without disturbing the stored information, so that the errors can be undone. This only helps if the physical qubits are good enough. Below a certain error rate per operation, the *threshold*, larger codes give lower logical error rates, as low as a computation requires, and the computer is called *fault tolerant*.

Fault tolerance is expensive. Useful algorithms are estimated to need of order a thousand physical qubits for every logical qubit, and around a million physical qubits in total (Fowler et al. 2012; Gidney 2025). Experiments are now crossing the threshold: in 2024, a superconducting device ran an error-correcting code whose logical error rate halved each time the code was made larger (Google Quantum AI and Collaborators 2025). Such experiments use a whole device to protect a single logical qubit, however, and useful computations still run on physical qubits. We come back to the stage in between, *early fault tolerance*, at the end of the course.

The cost of error correction is not only an increase in the number of physical qubits. A code protects stored information, but every gate on logical qubits must also be carried out without spreading errors, and only a small set of gates can be done this way at a low cost. Every other gate has to be built from this set, so the operations that are cheap on a NISQ device can be expensive on a fault-tolerant one, and the reverse. In the most studied codes, a CNOT between two logical qubits is relatively cheap. A single-qubit rotation by an arbitrary angle, which a NISQ device performs with a single pulse (Section 1.3.2), must instead be approximated by a sequence of tens to about a hundred protected gates, some of which need a costly extra procedure of their own (Ross and Selinger 2016; Litinski 2019). The cost of a NISQ circuit is dominated by its depth and its two-qubit gates, while the cost of a fault-tolerant circuit is often dominated by its number of rotations. Table 1.1 summarises the differences.

Table 1.1: NISQ and fault-tolerant quantum computing compared.

	NISQ	Fault tolerant
Qubits	physical qubits	logical qubits, each encoded in many physical qubits
Errors	uncorrected; they accumulate with every operation	corrected, if the physical error rates are below the threshold
Circuit size	limited by noise	limited by the number of qubits and the run time
Cheap operations	single-qubit rotations at any angle	CNOT and a few other gates
Expensive operations	two-qubit gates	single-qubit rotations at arbitrary angles
Cost of a circuit	mainly its depth and number of two-qubit gates	mainly its number of rotations

	NISQ	Fault tolerant
Dealing with noise	short circuits, many shots, error mitigation	error correction
Status	available today	first demonstrations; useful machines still to be built

This course focuses on NISQ devices. We will learn how noise affects their results, how error mitigation can reduce its effect, and which algorithms can be run on today's devices.

1.2 Physical qubits

This section is a high-level introduction to the most common designs of two kinds of hardware, superconducting circuits and trapped ions. Quantum hardware is a very active field of research, and there are many variations on everything we describe here. The device physics is descriptive, and you are not expected to derive it. What matters later are its consequences for gates, connectivity and speed. Terms we leave unexplained are in the glossary; on the web version, hover over an underlined term to see its definition.

1.2.1 Requirements for a quantum computer

A qubit is a quantum system with two states, $|0\rangle$ and $|1\rangle$, that we can control. Quantum systems such as atoms and small electrical circuits can only take certain values of energy, their *energy levels*, and they usually have many more than two. Most qubits are therefore the two lowest energy levels of a larger system. The other levels are still there, and a qubit can end up in them by accident.

A universal, *digital* quantum computer is built by assembling many physical qubits and controlling them individually. DiVincenzo listed what this requires (DiVincenzo 2000):

- a scalable system of well-characterised qubits: their levels and interactions are known, and more qubits can be added;
- a way to prepare the qubits in a known state, usually $|0\cdots0\rangle$;
- a universal set of gates, usually rotations of single qubits together with an interaction between two qubits;
- a way to measure individual qubits;
- *coherence times*, the times for which the qubits keep their quantum state, that are long compared with the duration of a gate.

These are the *DiVincenzo criteria*. Two further criteria concern sending qubits between distant devices, which we will not need.

One property of measurement matters well before Section 1.5, where readout is described in detail. On the platforms in this chapter, a measurement asks each qubit whether it is in $|0\rangle$ or in $|1\rangle$: it projects the qubit onto one of these two states and returns one bit. This is a measurement in the *computational basis*, also called the Z basis, since $|0\rangle$ and $|1\rangle$ are the eigenstates of Z. To measure in another basis, we first rotate that basis onto the computational one.

The requirements pull in opposite directions. A qubit keeps its state longest when it is well isolated from its surroundings, since every interaction with the environment disturbs it. Yet to apply gates and to measure, we have to interact with it, strongly and quickly. Every platform is a compromise between isolation and control, and this compromise explains many of the numbers in this chapter.

1.2.2 Energy levels, frequencies and pulses

Quantum mechanics links energy and frequency. An energy difference E between two levels corresponds to a frequency $f = E/h$, where h is Planck's constant, and radiation at that frequency drives transitions between the two levels, while radiation at other frequencies hardly affects them. This selectivity is called *resonance*, and on most platforms it is how qubits are controlled.

A *pulse* is a short burst of radiation at a chosen frequency, the carrier frequency, switched on and off smoothly. Its shape over time, the *envelope*, is often close to a Gaussian (Figure 1.1 a). Superconducting qubits are driven by microwave pulses, trapped ions by laser or microwave pulses. A qubit with transition frequency f_q oscillates at that frequency too, and in a simplified picture, the effect of a pulse on the qubit is the product of the pulse with this oscillation, added up over time. If the two frequencies are equal, the product keeps the same sign, and its effect builds up as long as the pulse lasts (Figure 1.1, left, b and c). The result is a rotation of the qubit, by an angle proportional to the area under the envelope. If the frequencies differ, the product alternates in sign, its contributions cancel, and the qubit is left almost unchanged. A pulse at the frequency of one qubit therefore acts on that qubit, and hardly on qubits with other frequencies. Section 1.3.2 shows how the shape and timing of a pulse set the rotation.

1.2.3 Superconducting qubits

A superconducting qubit is a small electrical circuit on a chip. The chip is *fabricated* like a conventional microchip: in a clean room, thin layers of superconducting metal are deposited on it and etched into shape (Figure 1.3). Once cooled, its metal becomes superconducting. Current then flows without resistance, so no energy is lost as heat, and the circuit behaves as a quantum system with discrete energy levels; we treat this part as a black box. A simple circuit

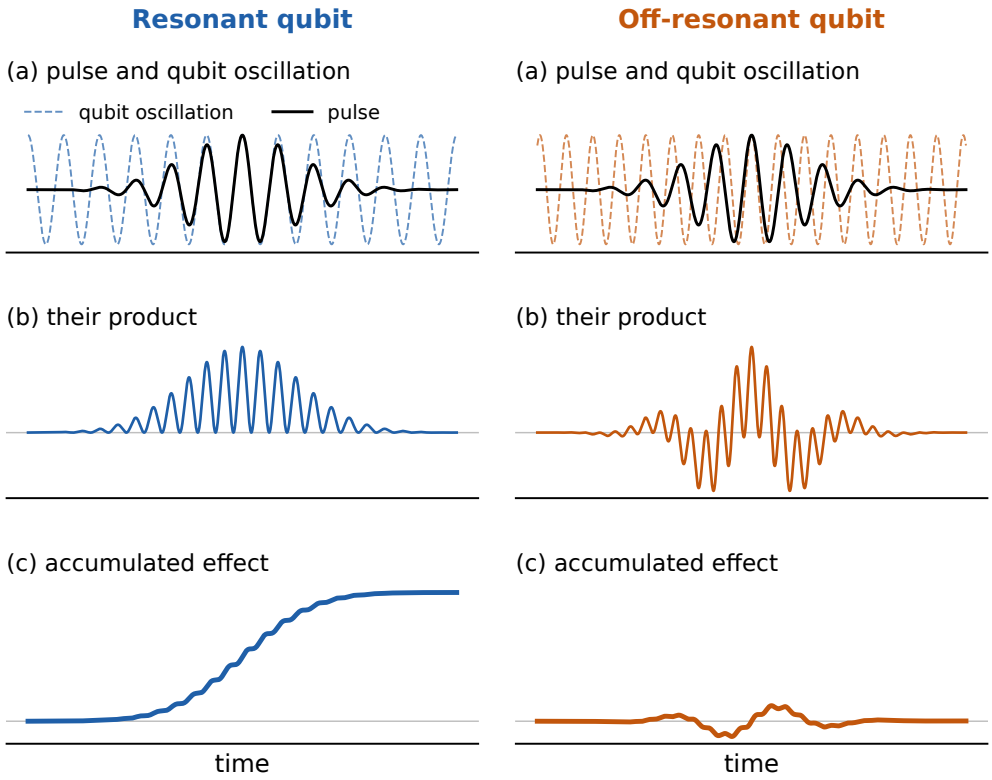


Figure 1.1: The same Gaussian pulse acting on a qubit at the carrier frequency (left) and on a qubit at a 35% higher frequency (right). (a) The pulse, with the qubit's own oscillation dashed; (b) the product of the two; (c) its running integral. The effect on the resonant qubit builds up to a value set by the area under the envelope; for the off-resonant qubit, it averages out to almost zero. The frequencies are scaled down to show the oscillations: a superconducting qubit oscillates a hundred times or more during a pulse, so that a much smaller difference in frequency is enough.

of this kind has evenly spaced levels, like a mass on a spring (Figure 1.2, left). That makes it useless as a qubit, because a pulse at the frequency f_{01} of the $|0\rangle \to |1\rangle$ transition is just as resonant with $|1\rangle \to |2\rangle$, $|2\rangle \to |3\rangle$ and so on. A circuit element called a *Josephson junction* makes the spacing uneven. In the most common design, the *transmon*, the $|1\rangle \to |2\rangle$ transition is a few percent lower in frequency than the qubit transition (Figure 1.2, right), so a pulse at f_{01} drives only the qubit.

Transmon frequencies are typically 4-6 GHz, in the microwave range. At room temperature, the thermal energy is about a thousand times larger than hf, so heat alone would excite the qubit at random. Superconducting chips therefore sit in a *dilution refrigerator* at about 10 mK, a hundredth of a degree above absolute zero (Exercise 1.1), and cables bring the microwave pulses down to them from electronics at room temperature.

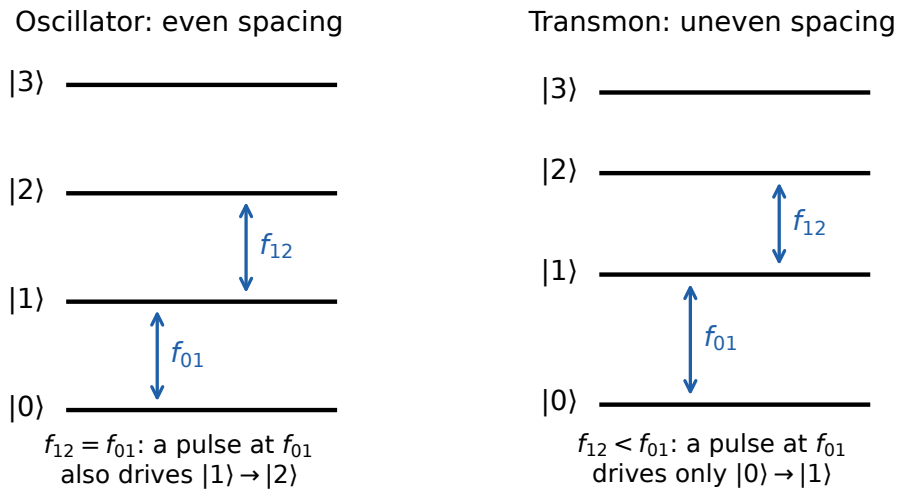
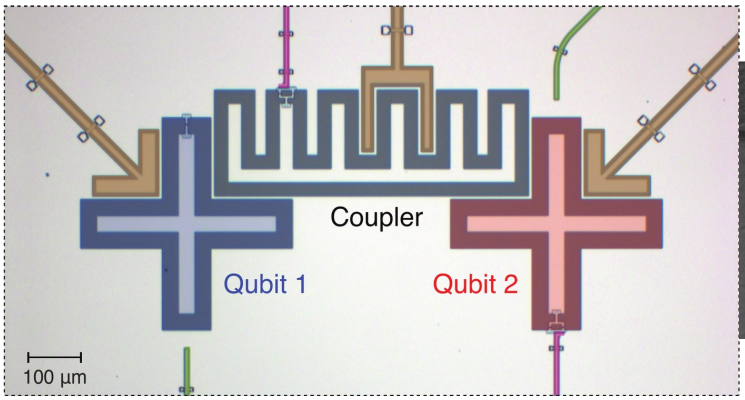


Figure 1.2: Energy levels of an oscillator and of a transmon. With even spacing, every transition has the same frequency, so the two lowest levels cannot be driven on their own. The transmon's uneven spacing makes the qubit transition unique. Not to scale: in a real transmon, f_{12} is only a few percent below f_{01}.

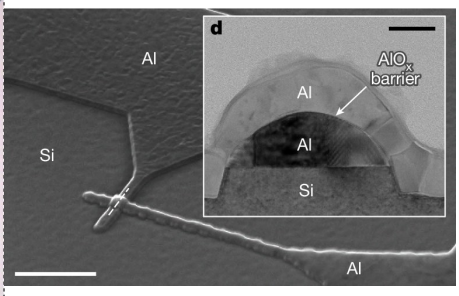
Superconducting qubits are fast. A transmon is large for a quantum system, a fraction of a millimetre across, and it behaves like a small antenna that couples strongly to microwaves, so a pulse of a few tens of nanoseconds is enough for a single-qubit gate. The same antenna would pick up any stray radiation, so the chip is shielded, inside a metal package and the nested shields of the refrigerator, and its control lines are heavily filtered. Ideally, these lines are the qubit's only contact with the outside world. The shielding is never perfect, and the quantum state of a qubit typically survives for tens to hundreds of microseconds.

Each qubit has its own frequency, and this is how the qubits on a chip are addressed separately. Every qubit usually has its own control line, but part of a pulse leaks to the neighbouring qubits, and coupled qubits feel each other's drive. Neighbouring qubits are therefore given different frequencies, so that a pulse meant for one of them is off resonance for the others (Figure 1.1). In many designs, the frequency can also be tuned. The Josephson junction is then replaced by a small loop with two junctions, and a magnetic field through the loop, set by the current in a nearby control line, changes the frequency of the qubit. Tuning lets the electronics move qubits towards each other in frequency and away again, which is one way to switch on a two-qubit gate.

Fabrication is never perfectly precise, so no two qubits come out identical. Each has its own frequency, its own coherence time and its own error rates. Every pulse must therefore be *calibrated*: its frequency, strength and duration are measured and adjusted, qubit by qubit and pair by pair, in a long sequence of automated experiments. The properties of the qubits also drift, over hours to days, so calibration is repeated regularly, and providers publish the



(a) Two transmons and a coupler.



(b) A Josephson junction.

Figure 1.3: Superconducting qubits under the microscope. (a) Two transmon qubits (blue and red, false colour) with a third transmon between them, used as a tunable coupler (grey) (Sung et al. 2021). The cross-shaped pads are the transmons' capacitors, and their Josephson junctions sit in the small structures at the ends of the arms. Qubit 2 and the coupler have two junctions in a loop, so that their frequencies can be tuned. Green lines carry the drive pulses, pink lines the currents that tune the frequencies, and the brown structures belong to the readout (Section 1.5.1). (b) A Josephson junction from a different device, seen in an electron microscope: two aluminium wires (Al) cross on a silicon chip (Si), separated by a thin oxide barrier, shown in cross-section in the inset (Van Damme et al. 2024). Scale bars: 1 µm; inset, 50 nm. Both images adapted (cropped) under the CC BY 4.0 licence.

resulting calibration data for each qubit (Section 1.4).

Two qubits can interact only if they are coupled, directly through a capacitor or through a small circuit between them, a *coupler* (Figure 1.3). The couplings are built into the chip, so which pairs can interact is fixed when the chip is made (Section 1.3.6). A coupling makes an interaction possible, and the electronics decide when it acts. Like a qubit and a pulse, two coupled qubits exchange energy efficiently only when their frequencies match. While their frequencies are kept apart, or the coupler is tuned to cancel the coupling, they hardly interact. A two-qubit gate switches the interaction on for a short time, by tuning the qubits or the coupler, or by driving one of the qubits (Section 1.3.2). The interaction never switches off completely, and what remains is a source of errors called *crosstalk*.

Exercise 1.1 (Thermal excitation)**.** In thermal equilibrium at temperature T, the excited and ground states of a qubit with frequency f are populated in the ratio $p_1/p_0 = e^{-hf/k_BT}$ (the Boltzmann factor), where k_B is Boltzmann's constant. Use $h/k_B \approx 0.048$ K/GHz.

(a) Compute p_1/p_0 for a 5 GHz qubit at room temperature (300 K), at 1 K and at 10 mK.

(b) Below which temperature is p_1/p_0 smaller than 10^{-3}?

Hint: Take the logarithm of both sides.

1.2.4 Trapped ions

A trapped-ion qubit is a single *ion*, an atom with one electron removed, typically of ytterbium, barium or calcium. The electrons of an atom can only be in certain states, each with its own energy. These are the atom's *internal* energy levels, as opposed to the motion of the atom as a whole, which will also matter below. Two of these levels form the qubit. All ions of an element are exactly the same, so, unlike fabricated circuits, the qubits of an ion-trap computer are identical by nature. As a consequence, they all have the same frequency, and a pulse cannot select one qubit by its frequency. Instead, most designs aim a laser beam at a single ion (Figure 1.5). In exchange, calibration is simpler, since all qubits share their frequency and most of their properties.

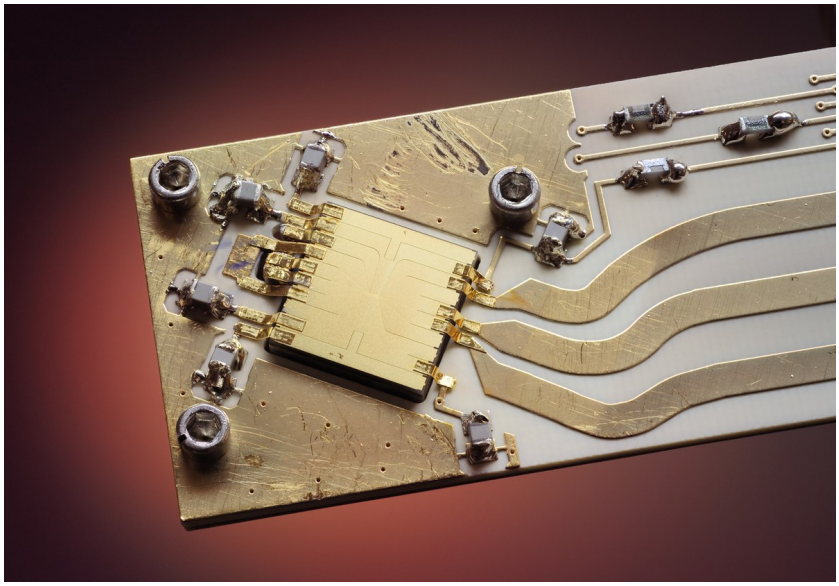
The ions are held in vacuum by electric fields in an *ion trap*, lined up in a chain a few micrometres apart, and laser beams slow them down until they are almost at rest (*laser cooling*). Laser beams, or in some designs microwaves, also drive the gates. The qubit levels can be chosen so that stray magnetic fields and other disturbances hardly affect them, and trapped-ion qubits keep their quantum state for a second or more, thousands of times longer than superconducting qubits. Figure 1.4 shows a trap chip and a chain of ions.

The ions are charged and repel each other, so the chain behaves like a row of balls joined by springs (Figure 1.5). Push one ion, and the whole chain starts to vibrate. Like the internal energy of an atom, the energy of these vibrations comes in discrete units, called *phonons*. Every ion takes part in the same vibrations, so they act as a shared channel, the *phonon bus*, through which any ion can interact with any other. The two-qubit gates are built by exploiting this phonon bus, so they can be applied to any pair of qubits: the connectivity is all-to-all (Section 1.3.6).

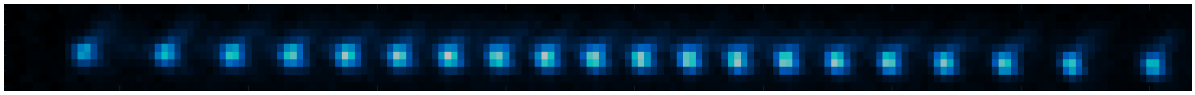
This comes at the cost of speed. Gates that use the phonon bus take from tens of microseconds to about a millisecond, roughly a thousand times longer than superconducting gates, and they become slower and harder to control as the chain grows. To scale up, some devices, such as Quantinuum's, divide the trap into zones and move ions between them, bringing the two ions of each gate together (*QCCD*).

1.2.5 Other platforms

Several other platforms are developing quickly, following the same recipe of a two-level system, a way to drive it, an interaction between qubits and a readout. In *neutral-atom* devices, uncharged atoms are held, and can be moved, by tightly focused laser beams (*optical tweezers*), and they interact when lasers excite them to very large *Rydberg states*; arrays of hundreds



(a) An ion-trap chip.



(b) A chain of twenty ions.

Figure 1.4: Trapped ions. (a) An ion trap made at NIST: a square gold chip, 7.4 mm on a side, on its circuit board. In experiments, two ions hover above the middle of the chip, and microwaves fed in through the three thick lines at the lower right drive their gates (photo: Y. Colombe/NIST, public domain). (b) Twenty calcium ions in a linear trap at Innsbruck, seen through the light they scatter: each bright spot is one ion, and the chain is about 0.1 mm long (Friis et al. 2018). Adapted (cropped) under the CC BY 4.0 licence.

to thousands of atoms have been built (Henriet et al. 2020). *Photonic* devices carry qubits on particles of light, which hardly interact with their surroundings but, for the same reason, hardly with each other either, so their entangling operations succeed only some of the time (Slussarenko and Pryde 2019). *Spin qubits* use the *spin* of single electrons trapped in tiny semiconductor structures (*quantum dots*); they can be made with the techniques of the chip industry, but current devices have at most a few tens of qubits (Burkard et al. 2023). Table 1.2 compares the five platforms.

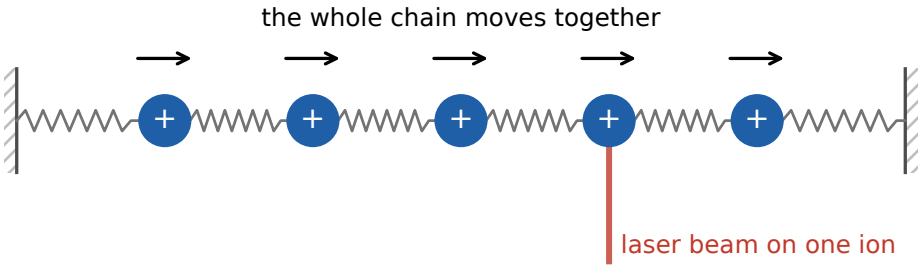


Figure 1.5: A trapped-ion chain as balls joined by springs. The ions repel each other, so they move together; shown here is the vibration in which the whole chain moves as one. A laser beam focused on a single ion drives that ion's qubit.

Table 1.2: Platforms for gate-based quantum computing, compared.

Platform	Qubit	Two qubits interact through	Strengths	Main limitations
Superconducting	energy levels of a circuit on a chip	couplings built into the chip	fast gates; mature fabrication technology	short coherence; fixed connectivity; qubits not identical
Trapped ions	energy levels of an ion	shared vibrations of the chain	identical qubits; long coherence; any pair can interact	slow gates; long chains are hard to control
Neutral atoms	energy levels of an atom	excitation to large Rydberg states	many qubits; atoms can be rearranged	slow; atoms get lost and must be replaced
Photonics	a photon	interference and detection	little decoherence; photons travel far	entangling operations often fail; photons get lost
Spin qubits	spin of an electron	neighbouring electrons	very small; chip-industry fabrication	few qubits so far; qubits not identical

1.3 Native gates and connectivity

The circuits we write can use any gate that is convenient. A device, however, implements only a small set of *native gates*, each one a calibrated pulse or a short sequence of pulses, and it applies two-qubit gates only between pairs of qubits that are physically coupled. An additional step, compilation, translates every circuit into these native operations. This section explains which gates are native and why, and what the translation costs.

1.3.1 Pauli rotations

Recall the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are Hermitian and square to the identity, $X^2 = Y^2 = Z^2 = I$. Any two different Pauli matrices anticommute, for example $XZ = -ZX$, and the product of two of them is the third, up to a factor: $XY = iZ$, $YZ = iX$ and $ZX = iY$. On several qubits, we combine them with tensor products.

Definition 1.1 (Pauli string)**.** A *Pauli string* on n qubits is a tensor product $P = P_1 \otimes P_2 \otimes \cdots \otimes P_n$ with each $P_j \in \{I, X, Y, Z\}$. We often drop the tensor product signs and write, for example, ZZ or XIY.

A Pauli string is again Hermitian, and it squares to the identity, because the square of a tensor product is the tensor product of the squares: $P^2 = P_1^2 \otimes \cdots \otimes P_n^2 = I$. This one property gives a closed formula for the exponential of P. Expand the exponential as a power series. Even powers of P are the identity, $P^{2m} = I$, and odd powers are $P^{2m+1} = P$, so the series splits into two:

$$e^{-i\theta P/2} = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} \left(\frac{\theta}{2}\right)^{2m} I \;-\; i \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} \left(\frac{\theta}{2}\right)^{2m+1} P.$$

These are the power series of the cosine and the sine, so

$$e^{-i\theta P/2} = \cos\frac{\theta}{2}\, I - i \sin\frac{\theta}{2}\, P. \tag{1.1}$$

The derivation used only $P^2 = I$, so Equation 1.1 holds for any operator with that property. Such operators are called *involutory*. For Hermitian operators, like all those in this chapter, being involutory is the same as being a *reflection*: the eigenvalues are ± 1, so the operator flips the sign of one subspace and leaves the orthogonal subspace unchanged.

Applied to single Pauli matrices, Equation 1.1 gives the familiar rotations of a qubit about the axes of the Bloch sphere, for example

$$R_X(\theta) = e^{-i\theta X/2} = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \qquad R_Z(\theta) = e^{-i\theta Z/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}. \tag{1.2}$$

Rotations about two different axes are enough to build any single-qubit gate. Up to a global phase, every single-qubit unitary can be written as $R_Z(\alpha)\, R_Y(\beta)\, R_Z(\gamma)$ for some angles (Nielsen and Chuang 2010, sec. 4.2).

Applied to the two-qubit Pauli string $Z \otimes Z$, Equation 1.1 gives

$$R_{ZZ}(\theta) = e^{-i\theta\, Z\otimes Z/2} = \begin{pmatrix} e^{-i\theta/2} & 0 & 0 & 0 \\ 0 & e^{i\theta/2} & 0 & 0 \\ 0 & 0 & e^{i\theta/2} & 0 \\ 0 & 0 & 0 & e^{-i\theta/2} \end{pmatrix} \tag{1.3}$$

in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. R_{ZZ} applies a phase that depends only on whether the two bits are equal or different, their parity. That is why it can be built from two CNOTs and a single R_Z (Figure 1.6). The first CNOT writes the parity of the two bits into the second qubit, R_Z applies the phase, and the second CNOT undoes the first. Section 1.3.4 extends this construction to rotations about any Pauli string, built from native gates.

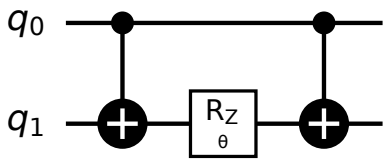


Figure 1.6: $R_{ZZ}(\theta)$ built from two CNOTs and one $R_Z(\theta)$.

Equation 1.1 will come back several times in this course, in circuits for time evolution, which are built from Pauli rotations, and in the parameter-shift rule, which gives exact gradients of variational circuits.

1.3.2 Physical implementation of gates

The native gates of a qubit platform are set by the physical mechanisms that control its qubits. This section describes these mechanisms, for single-qubit and for two-qubit gates, and the gates they produce.

A single-qubit gate on a superconducting qubit is a microwave pulse at the qubit's frequency, a few tens of nanoseconds long (Section 1.2.2). The area under the envelope of the pulse, set by its strength and duration, sets the rotation angle θ (Figure 1.1). The phase of the pulse, that is, the timing of the microwave oscillation relative to a reference clock in the control electronics, sets the axis of rotation, which lies in the xy-plane of the Bloch sphere at an angle φ from the x-axis. Delaying the oscillation by a fraction of its period turns the axis by the same fraction

of a full turn. Laser or microwave pulses on trapped ions work in the same way. On both platforms, the natural single-qubit gate is therefore a rotation by any angle about any axis in the xy-plane, which IQM calls PRX:

$$\mathrm{PRX}(\theta, \varphi) = \exp\left[-i\frac{\theta}{2}\left(\cos\varphi\, X + \sin\varphi\, Y\right)\right]. \tag{1.4}$$

The operator $\cos\varphi\, X + \sin\varphi\, Y$ squares to the identity (Exercise 1.2), so Equation 1.1 applies to it too. For $\varphi = 0$ and $\varphi = \pi/2$, PRX reduces to $R_X(\theta)$ and $R_Y(\theta)$. Figure 1.7 shows how PRX rotates the Bloch sphere. Rotations about the z-axis are missing from this family, and Section 1.3.5 shows that they need no pulse at all.

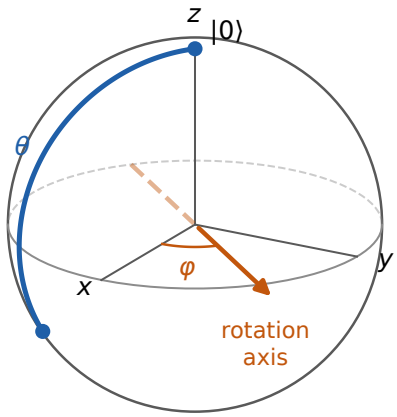


Figure 1.7: The Bloch sphere rotated by $\mathrm{PRX}(\theta, \varphi)$, a rotation by θ about the axis in the xy-plane at an angle φ from the x-axis. The blue path shows where $|0\rangle$ goes as the rotation angle grows from 0 to θ. In the web version, the figure is interactive: the sliders set θ and φ, the blue grid turns with the sphere, and dragging changes the view.

Exercise 1.2 (The PRX gate)**.**

(a) Show that $(\cos\varphi\, X + \sin\varphi\, Y)^2 = I$.

Hint: Expand the square, and use that X and Y anticommute.

(b) Write $\mathrm{PRX}(\theta, \varphi)$ as a 2×2 matrix.

Hint: Use Equation 1.1, which applies because of (a).

(c) Check that $\mathrm{PRX}(\theta, 0) = R_X(\theta)$ and $\mathrm{PRX}(\theta, \pi/2) = R_Y(\theta)$. Which rotation is $\mathrm{PRX}(\theta, \pi)$?

During a two-qubit gate, the two qubits should interact, and at all other times they should not (Section 1.2.3). Superconducting devices switch the interaction on in three main ways, and each gives a different native gate.

- **Tuning two qubits into resonance.** When two coupled qubits have the same frequency, an excitation moves back and forth between them: $|01\rangle$ turns into $|10\rangle$ and back again, while $|00\rangle$ and $|11\rangle$ are unchanged. A complete transfer gives the iSWAP gate, which exchanges $|01\rangle$ and $|10\rangle$ and multiplies both by i. Stopping halfway gives its square root, $\sqrt{\text{iSWAP}}$.
- **Using the third level.** A transmon has a third level, $|2\rangle$, not far above the qubit levels (Figure 1.2). The frequencies can be tuned so that the two-qubit state $|11\rangle$ has the same energy as $|20\rangle$, in which the first transmon holds both excitations. These two states then exchange instead. After one complete exchange, there and back, $|11\rangle$ returns with its sign flipped (up to phases on the single qubits, which are easy to correct), while $|00\rangle$, $|01\rangle$ and $|10\rangle$, which have no partner at the same energy, are unchanged. This is the controlled-Z gate, CZ, which flips the sign of $|11\rangle$ only. On most current devices, a tunable coupler between the qubits switches the interaction, and CZ is the most common native gate.
- **Driving one qubit at its neighbour's frequency.** IBM's older Eagle devices, retired in 2026, have qubits with fixed frequencies and couplings that are always on. A pulse on one qubit at the frequency of its neighbour (*cross-resonance*) makes the neighbour rotate about its x-axis, in a direction that depends on the state of the first qubit: a rotation generated by $Z \otimes X$. In practice, it is applied as a gate called ECR, for echoed cross-resonance, which equals a CNOT up to single-qubit gates.

On trapped ions, laser beams on two ions couple their qubit states to the shared vibrations of the chain, the phonon bus of Section 1.2.4. The light pushes each ion with a force that depends on its qubit state, so the chain vibrates in a way that depends on both qubits, and through this vibration the two qubits interact. At the end of the gate the vibration returns to its starting state, so that it carries away no information about the qubits. This *Mølmer-Sørensen gate* is a rotation generated by $X \otimes X$, of the form of Equation 1.1, with an angle set by the duration and strength of the laser pulses. It is available at any angle, from small rotations up to a maximally entangling one, and between any pair of ions in the chain. Combined with single-qubit rotations, it gives $R_{ZZ}(\theta)$ at any angle, the native two-qubit gate of current trapped-ion devices.

1.3.3 Native gate sets

Table 1.3 lists the native gates of some current devices. Names and details change with each hardware generation, but the pattern stays the same: a continuous family of single-qubit rotations, and one or a few entangling gates. Together they are universal, so any circuit can be compiled into them, at the cost of extra gates.

Table 1.3: Native gates of some current devices, as of 2026. All of them also offer virtual Z rotations (Section 1.3.5).

Device family	Single-qubit	Two-qubit
IQM (superconducting)	PRX(θ, φ)	CZ; on star devices, also MOVE to and from the central resonator
IBM (superconducting)	$\sqrt{X}$, X; experimental: $R_X(\theta)$	CZ (Heron, Nighthawk); ECR on the retired Eagle devices; experimental: $R_{ZZ}(\theta)$
Google (superconducting)	PhasedXZ	CZ (Willow); $\sqrt{\text{iSWAP}}$ and others on the older Sycamore
Quantinuum (trapped ions)	$R_{xy}(\theta, \varphi)$	$R_{ZZ}(\theta)$, any angle
IonQ (trapped ions)	GPi(φ), GPi2(φ)	MS (Aria), ZZ(θ) (Forte), any angle

The single-qubit gates in the table are rotations about axes in the xy-plane under different names, in one case followed by a Z rotation:

- $\sqrt{X}$ and X (IBM) are $R_X(\pi/2)$ and $R_X(\pi)$, up to a global phase; IBM's Heron devices also offer $R_X(\theta)$ at any angle, as an experimental option;
- PhasedXZ (Google) is an xy-plane rotation followed by a Z rotation;
- $R_{xy}(\theta, \varphi)$ (Quantinuum, also called U1q) is PRX(θ, φ) under another name;
- GPi(φ) and GPi2(φ) (IonQ) are PRX(π, φ) and PRX$(\pi/2, \varphi)$, up to a global phase. IonQ measures angles in *turns*, so that an angle of 1 is a full turn, 2π.

The two-qubit gates are either fixed gates or rotations with an adjustable angle. The fixed ones are CZ, $\sqrt{\text{iSWAP}}$ and ECR. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\text{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad \sqrt{\text{iSWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and ECR equals a CNOT up to single-qubit gates on both qubits. The adjustable ones are $R_{ZZ}(\theta)$ of Equation 1.3, IonQ's ZZ(θ), which is the same gate with its angle in turns, and the Mølmer-Sørensen gate MS, generated by $X \otimes X$. With an adjustable angle, an entangling rotation of any strength is a single native gate, which saves gates in circuits built from Pauli rotations.

1.3.4 Compiling into native gates

Compiling a circuit into native gates relies on an identity that we will use several times: conjugating an exponential conjugates its exponent,

$$U e^{A} U^{\dagger} = e^{UAU^{\dagger}} \tag{1.5}$$

for any unitary U and any square matrix A. To see this, expand both sides as power series: the inner factors $U^{\dagger}U = I$ cancel in $(UAU^{\dagger})^{k} = UA^{k}U^{\dagger}$. For a Pauli rotation, Equation 1.5 says that $Ue^{-i\theta P/2}U^{\dagger} = e^{-i\theta UPU^{\dagger}/2}$: conjugated by U, a rotation about P becomes a rotation about $UPU^{\dagger}$, by the same angle. To rotate about a long Pauli string, it is therefore enough to find a circuit U of native gates that turns a simple Pauli matrix, such as X on a single qubit, into that string.

Two kinds of conjugation do the job. First, single-qubit rotations turn one Pauli matrix into another; for example, $R_Y(\pi/2)$ turns Z into X, and $R_X(-\pi/2)$ turns Z into Y (Exercise 1.3). Second, CZ attaches a Z to an X on the other qubit:

$$\mathrm{CZ}\,(I \otimes X)\,\mathrm{CZ} = Z \otimes X.$$

To check this, apply both sides to a basis state $|ab\rangle$. On the left, the first CZ gives a sign $(-1)^{ab}$, X flips b, and the second CZ gives a sign $(-1)^{a(1-b)}$, so the result is $(-1)^{a}\,|a, 1-b\rangle$, which is the action of $Z \otimes X$. Each further CZ on the same qubit attaches one more Z: writing CZ_{jk} for a CZ on qubits j and k, and labelling single-qubit Paulis by their qubit, conjugation by $\mathrm{CZ}_{13}\mathrm{CZ}_{23}$ turns X_3 into $Z_1 Z_2 X_3$. Single-qubit conjugations then turn each factor into the Pauli matrix we need. Figure 1.8 shows the result for $P = Z \otimes Y \otimes X$. Every gate in it is native on a device with PRX and CZ, since $R_X(\alpha) = \mathrm{PRX}(\alpha, 0)$.

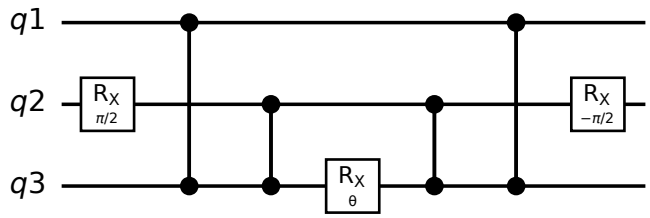


Figure 1.8: $e^{-i\theta\, Z\otimes Y\otimes X/2}$ built from CZ and R_X rotations. The CZs turn the X rotation on q_3 into a rotation about $Z \otimes Z \otimes X$, and the outer R_X rotations on q_2 turn its Z into Y.

A rotation about a Pauli string on k qubits costs $2(k-1)$ CZs this way, the same number of two-qubit gates as the CNOT construction of Figure 1.6. In Figure 1.8, all CZs act on the qubit that carries the rotation, which suits a device where that qubit is coupled to all the others. On a line of qubits, a chain of CNOTs between neighbours collects the parity step by step instead, and each CNOT is one CZ between two single-qubit rotations (Exercise 1.6).

Exercise 1.3 (Turning one Pauli into another)**.** Let P and Q be Pauli strings that anticommute, $PQ = -QP$.

(a) Show that $e^{-i\alpha P/2}\, Q\, e^{i\alpha P/2} = \cos\alpha\, Q - i \sin\alpha\, PQ$.

Hint: Write both exponentials with Equation 1.1, and use $PQP = -Q$.

(b) Show that $R_Y(\pi/2)$ turns Z into X, and that $R_X(-\pi/2)$ turns Z into Y.

Hint: Use (a), and the products of Pauli matrices.

(c) Check that the circuit of Figure 1.8 implements $e^{-i\theta\, Z\otimes Y\otimes X/2}$.

Hint: Use Equation 1.5, the identity for CZ in the text, and (b).

(d) Draw a circuit of CZ gates and R_X or R_Y rotations for $R_{ZZ}(\theta)$, and compare it with Figure 1.6.

Hint: Start from $R_X(\theta)$ on the second qubit, conjugate it by CZ, then turn the remaining X into Z.

1.3.5 Virtual Z gates

PRX contains no rotations about the z-axis, yet circuits are full of them, for example in the decomposition $R_Z(\alpha)\, R_Y(\beta)\, R_Z(\gamma)$ of a general single-qubit gate (Section 1.3.1) and in many parametrised gates of algorithms. On most devices, superconducting and trapped-ion alike, Z rotations are implemented without any pulse, as *virtual Z gates*. The trick follows from Pauli algebra.

Conjugation by a Z rotation rotates X and Y within the xy-plane:

$$R_Z(\alpha)\, X\, R_Z(\alpha)^\dagger = \cos\alpha\, X + \sin\alpha\, Y, \qquad R_Z(\alpha)\, Y R_Z(\alpha)^\dagger = \cos\alpha\, Y - \sin\alpha\, X$$

(Exercise 1.4). By the angle-addition formulas, it therefore maps $\cos\varphi\, X + \sin\varphi\, Y$ to $\cos(\varphi + \alpha)\, X + \sin(\varphi+\alpha)\, Y$. By Equation 1.5, conjugating a PRX pulse by a Z rotation therefore shifts its phase:

$$R_Z(\alpha)\, \mathrm{PRX}(\theta, \varphi)\, R_Z(\alpha)^\dagger = \mathrm{PRX}(\theta, \varphi + \alpha). \tag{1.6}$$

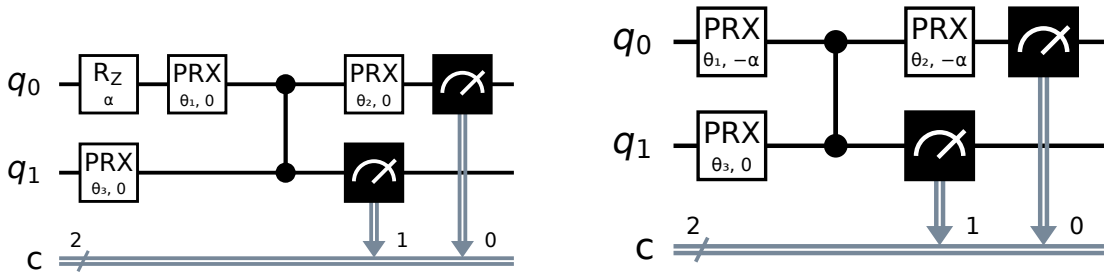
Replacing α by $-\alpha$ and multiplying by $R_Z(\alpha)$ on the left gives

$$\mathrm{PRX}(\theta, \varphi)\, R_Z(\alpha) = R_Z(\alpha)\, \mathrm{PRX}(\theta, \varphi - \alpha). \tag{1.7}$$

Operators act from right to left, so on the left-hand side $R_Z(\alpha)$ acts first and the pulse second. On the right-hand side, the pulse acts first, with its phase shifted by $-\alpha$. A Z rotation can therefore be moved later in a circuit, past a pulse, by changing the pulse's phase.

Now take a circuit with a Z rotation on one qubit (Figure 1.9a), and move the Z rotation forward in time. Each PRX it passes gets its phase shifted by $-\alpha$. CZ is diagonal, like R_Z, so

the two commute and the Z rotation passes through unchanged. At the end, the Z rotation sits just before a measurement in the computational basis. A measurement only asks whether each qubit is in $|0\rangle$ or in $|1\rangle$ (Section 1.2.1), and the Z rotation only changes the phases of $|0\rangle$ and $|1\rangle$, not the probabilities of measuring them, so it can be dropped. The result (Figure 1.9b) contains no Z rotation at all.



(a) With a Z rotation on q_0.

(b) The Z rotation absorbed into the phases of the later pulses on q_0.

Figure 1.9: Two circuits that give the same measurement statistics.

In hardware, the phase of each pulse, microwave or laser, is measured against a reference clock in the control electronics, and a virtual Z gate shifts that reference for all later pulses on the qubit. It is an instruction to the electronics, carried out instantly and exactly, so Z rotations take no time and add no error. When we estimate the error of a circuit, only the other gates count.

The same trick makes every single-qubit gate cost a single pulse. Write the gate as $R_Z(\alpha)\,R_Y(\beta)\,R_Z(\gamma)$, note that $R_Y(\beta) = \mathrm{PRX}(\beta, \pi/2)$, and move $R_Z(\gamma)$ past the pulse with Equation 1.7:

$$R_Z(\alpha)\,R_Y(\beta)\,R_Z(\gamma) = R_Z(\alpha+\gamma)\,\mathrm{PRX}(\beta, \pi/2-\gamma).$$

That is one pulse, followed by a Z rotation that is virtual.

Exercise 1.4 (Virtual Z gates)**.**

(a) Show that $R_Z(\alpha)\,X\,R_Z(\alpha)^\dagger = \cos\alpha\,X + \sin\alpha\,Y$ and $R_Z(\alpha)\,Y R_Z(\alpha)^\dagger = \cos\alpha\,Y - \sin\alpha\,X$.

Hint: Use Exercise 1.3 (a) with $P = Z$.

(b) Write the Hadamard gate, up to a global phase, as a single PRX pulse followed by a Z rotation.

Hint: Write the Hadamard gate as $R_Z(\alpha)\,R_Y(\beta)\,R_Z(\gamma)$ first, then use Equation 1.7.

1.3.6 Connectivity

A native two-qubit gate is available only between qubits that are physically coupled. The *coupling graph* of a device records which pairs these are. Its vertices are the qubits, and an edge joins two qubits if a native two-qubit gate can act on them (Figure 1.10).

On a superconducting chip, the coupling graph is fixed when the chip is made. In a square lattice, used by Google, IQM and IBM's newest devices, each qubit has up to four neighbours. IBM's heavy-hex lattice, used on most of its current devices, gives each qubit only two or three, which reduces unwanted interactions between qubits but lengthens the routes between distant ones. Some IQM devices use a star layout, in which every qubit couples to a central resonator, a circuit element that can take over the state of one qubit and pass it on to another; in Figure 1.10 it is drawn as an extra vertex.

In a trapped-ion chain, every ion couples to the same phonon bus, so every pair can interact and the coupling graph is complete. Devices that move ions between zones achieve the same by bringing any two ions together.

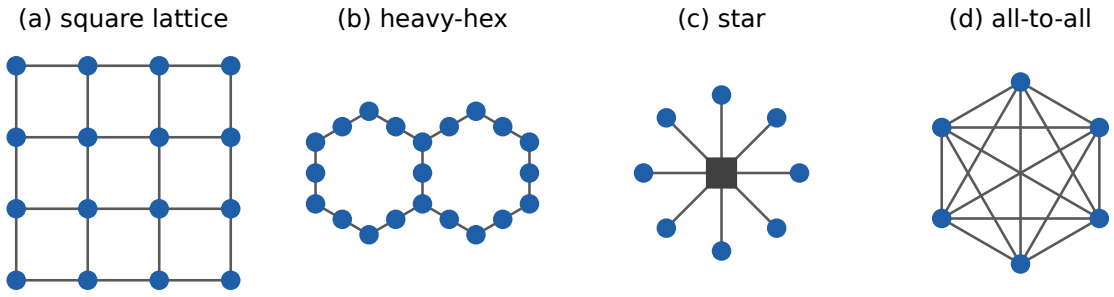


Figure 1.10: Coupling graphs: (a) square lattice; (b) heavy-hex lattice; (c) star, with every qubit coupled to a central resonator (square); (d) all-to-all, as in a trapped-ion chain.

What if a circuit needs a two-qubit gate between qubits that are not neighbours? Their states must first be brought together with *SWAP* gates, which exchange the states of two neighbouring qubits, and each SWAP costs three CNOTs (Figure 1.11). A gate between distant qubits can therefore cost many extra two-qubit gates (Exercise 1.5). Two-qubit gates are the main source of error on current devices (Section 1.4), so connectivity directly limits which circuits a device can run reliably.

In practice, nobody inserts SWAPs by hand. *Transpilation* maps a circuit onto the native gates and coupling graph of a device. It chooses which physical qubit plays the role of each qubit in the circuit, inserts SWAPs where needed, and simplifies the result. We return to its cost in later chapters.

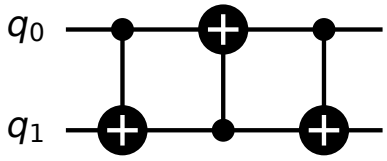


Figure 1.11: SWAP as three CNOTs with alternating control.

Exercise 1.5 (The cost of distance)**.**

(a) Check Figure 1.11 on the four computational basis states.

(b) We want to implement a CNOT between the qubits at opposite ends of a line of n qubits. If the qubits may stay in their new positions after the operation, how many CNOTs are needed? And if they must be returned to their original positions?

Hint: Each SWAP moves a state one step along the line.

Exercise 1.6 (SWAP from native gates)**.** On a device whose native gates are PRX and CZ, a SWAP must be built from these.

(a) Let A be a single-qubit unitary. Show that $(I \otimes A)\,\mathrm{CZ}\,(I \otimes A^\dagger)$ applies $AZA^\dagger$ to the second qubit if the first is in $|1\rangle$, and does nothing if it is in $|0\rangle$.

Hint: Write $\mathrm{CZ} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Z$.

(b) Find a PRX rotation A with $AZA^\dagger = X$, and write a CNOT as one CZ and two PRX gates.

Hint: Exercise 1.3 (b) gives a suitable A.

(c) Using Figure 1.11, write a SWAP with three CZs and PRX gates only. How many PRX pulses does it need?

1.4 Sources of noise

Every wire or laser beam used to control a qubit is also a way in for disturbances. This section describes in words what can go wrong; the next chapter goes deeper into the mathematical description of noise and its effects.

The most important effect is that an excited qubit loses its energy. Left alone, $|1\rangle$ decays to $|0\rangle$,

a process called *relaxation*. In superconducting qubits, the energy is absorbed by tiny defects in the materials of the chip. The qubit levels of trapped ions hardly decay on their own, but the lasers used for gates occasionally knock an ion out of its qubit state.

A qubit can also lose its phase without losing energy. A superposition of $|0\rangle$ and $|1\rangle$ carries a relative phase between its two parts, and the qubit's frequency sets how fast this phase turns. If the frequency fluctuates, for example because of fluctuating magnetic fields or electric charges near the qubit, the phase wanders, and the superposition gradually becomes a random mixture of $|0\rangle$ and $|1\rangle$. This is called *dephasing*. The time over which a qubit relaxes is called T_1, and the time over which it dephases is called T_2. These are the coherence times quoted for every device, and the next chapter defines them precisely.

A third kind of error is a gate that rotates by the wrong amount, because a pulse is slightly too long, too short or off axis, or because a laser's intensity fluctuates. Unlike relaxation and dephasing, such *coherent errors* are systematic: the same gate makes the same mistake every time. Table 1.4 lists these effects together with the remaining ones.

Table 1.4: What can go wrong in a quantum device.

What happens	Example	Name in the next chapter
$\|1\rangle$ decays to $\|0\rangle$	energy absorbed by defects in the chip	relaxation (T_1)
the relative phase of a superposition wanders	fluctuating magnetic fields	dephasing (T_2)
a gate rotates by the wrong amount	a pulse slightly too long	coherent error
the qubit leaves the qubit levels	a pulse pushes a transmon into its third level (Figure 1.2)	*leakage*
an operation on one qubit disturbs another	couplings that are never fully off; a laser beam that partly hits the next ion	crosstalk
the readout is wrong	0 reported as 1, or the reverse	readout error
many qubits fail at once	a cosmic ray deposits energy in the chip and disturbs all qubits nearby	correlated errors
the device drifts	properties that change over hours	(recalibration)

The last row explains why providers recalibrate their devices regularly and publish calibration data, the error rates of each qubit and gate at the last calibration. These data are a snapshot, and a few hours later the device may behave differently.

Providers summarise the quality of each operation by a *fidelity*. For now, read it as one minus the probability that the operation goes wrong; the next chapter gives the precise definition. Typical values on current devices are 99.9-99.99% for single-qubit gates, 99-99.9% for two-qubit gates and 96-99.9% for readout, so two-qubit gates and readout are the weakest links.

1.5 Measurement

1.5.1 Readout

On superconducting devices, each qubit is coupled to a small resonant circuit, its readout resonator, whose frequency shifts slightly depending on whether the qubit is in $|0\rangle$ or $|1\rangle$. To measure, the electronics send a weak microwave pulse to the resonator and decide between 0 and 1 from the signal that comes back (*dispersive readout*). This takes about a microsecond. On trapped ions, a laser beam is tuned so that an ion in one qubit state scatters light and appears bright, while an ion in the other state stays dark. A camera or photon detectors count the light from each ion, and this fluorescence readout takes from a hundred microseconds to a millisecond.

On both platforms, a measurement returns one bit per qubit, in the computational basis (Section 1.2.1). It projects each qubit onto $|0\rangle$ or $|1\rangle$, and measuring in another basis means first rotating that basis onto the computational basis.

1.5.2 Shots and expectation values

A single run of a circuit, from preparing the qubits to measuring them, is called a *shot*. It returns one bit string, drawn at random from the distribution given by the Born rule, so to learn anything quantitative we repeat the circuit many times.

Most quantities in this course are expectation values $\langle P\rangle$ of Pauli strings. To estimate one, we rotate each qubit so that its factor of P becomes Z (nothing for Z, a Hadamard for X, and $R_X(\pi/2)$ for Y), and measure. A Z measurement with outcome bit b corresponds to the eigenvalue $(-1)^b$ of Z, so the outcome of P is the product of these signs over the qubits on which P acts:

$$s = (-1)^{\sum_{j:\,P_j\neq I} b_j} = \pm 1.$$

For $P = ZZ$, for example, $s = +1$ if the two bits agree and $s = -1$ if they differ. The average of s over N shots,

$$\bar{s} = \frac{1}{N}\sum_{k=1}^{N} s_k, \tag{1.8}$$

is our estimate of $\langle P\rangle$. A rule like this, which turns data into an estimate, is called an *estimator*.

1.5.3 Shot noise

How many shots does an estimate need? Each shot gives $s = +1$ or $s = -1$ at random, and the expectation value of s is $\langle P \rangle$. The sample mean $\bar{s}$ of Equation 1.8 is therefore correct on average, or *unbiased*. Its random error comes from the spread of single shots. Since $s^2 = 1$ for every shot (this is $P^2 = I$ again), the variance of a single shot is

$$\mathrm{Var}(s) = \langle s^2 \rangle - \langle s \rangle^2 = 1 - \langle P \rangle^2.$$

The mean of N independent samples has $1/N$ times the variance of a single sample (for a refresher on this and other basic statistics, see Blitzstein and Hwang (2019)). The typical size of the random error of $\bar{s}$, its *standard error*, is therefore

$$\sigma_{\bar{s}} = \sqrt{\frac{1 - \langle P \rangle^2}{N}}. \tag{1.9}$$

In practice $\langle P \rangle$ is unknown, and we evaluate Equation 1.9 with $\bar{s}$ in its place.

The error is largest when $\langle P \rangle = 0$, where every shot is a coin flip, and vanishes when $\langle P \rangle = \pm 1$, where every shot gives the same result. It decreases only as $1/\sqrt{N}$, so each extra digit of precision costs a hundred times more shots. To estimate an expectation value near zero with a standard error of 0.01, for example, we need $N = 1/0.01^2 = 10^4$ shots. Section 1.5.4 shows how long they take on each platform.

Shot noise is one of two kinds of error in an estimated expectation value. Noise in the device also shifts $\langle P \rangle$ itself, usually towards zero, and more shots cannot remove that shift; they only estimate the shifted value more precisely. The next chapter separates these two kinds of error. Tutorial 1 practises Equation 1.8 and Equation 1.9 in code: open the T1 notebook in Colab.

1.5.4 Timescales

Table 1.5 collects typical timescales of current devices. They vary between devices and improve every year, so read them as orders of magnitude.

Table 1.5: Typical timescales of current devices, as orders of magnitude.

Time per	Superconducting	Trapped ions
single-qubit gate	20-60 ns	10-150 µs
two-qubit gate	20-700 ns	50 µs-1 ms
readout	0.5-2.5 µs	0.1-1 ms
coherence (T_2)	20-350 µs	about 1 s or more
shot (reset, circuit, readout)	0.3-4 ms	10 ms to several seconds
switching to a new circuit	much longer than a shot, plus any queue	the same

The coherence time divided by the two-qubit gate time is between about a thousand and ten thousand on both platforms, although their speeds differ by a factor of about a thousand. Both platforms thus meet DiVincenzo's requirement of coherence times that are long compared with gate times (Section 1.2.1), by a similar margin and in opposite ways: superconducting qubits with fast gates, trapped ions with long coherence. Coherence alone would allow circuits of thousands of gates in sequence. In practice, gate errors of 0.1-1% per two-qubit gate limit useful circuits much earlier.

Where the time goes differs between the platforms. On superconducting devices, the gates are fast, and most of a shot is spent measuring the qubits and resetting them to $|0\rangle$; devices run hundreds to thousands of shots per second. Switching to a different circuit costs much more, since the circuit must be compiled and loaded into the control electronics, and on a shared cloud device it may wait in a queue. Many shots of one circuit are cheap; many different circuits are expensive. On trapped ions, the gates themselves are slow, and each shot also includes cooling the ions and, on some devices, moving them between zones, which takes most of the time. There, the number of shots is the main cost. For the 10^4 shots of the example in Section 1.5.3, this means between a few seconds and a minute on a superconducting device, and from a few minutes to several hours on an ion trap.

Exercise 1.7 (A shot budget)**.**

(a) How many shots give a standard error of 0.01 when $\langle P\rangle \approx 0.9$? Compare with $\langle P\rangle \approx 0$.

Hint: Solve Equation 1.9 for N.

(b) You need ten expectation values, each from its own circuit, each with a standard error of at most 0.005. Assume the worst case, $\langle P\rangle = 0$. Estimate the total time on a superconducting device and on an ion trap. Which part of the time dominates on each?

Hint: Use the times per shot and per circuit of Table 1.5.

1.6 More exercises

These exercises combine material from several sections of the chapter.

Exercise 1.8 (An error budget)**.**
A device has the native gates PRX and CZ, with virtual Z rotations. Its operations fail independently, with probability 10^{-4} for a PRX pulse, 5×10^{-3} for a CZ and 10^{-2} for a measurement. On a line of 10 qubits, we run a brick-wall circuit: each layer applies R_Z and R_X to every qubit, followed by CZs between neighbouring qubits, on the pairs (1,2),

(3,4), … in odd layers and (2,3), (4,5), … in even layers. Figure 1.12 shows two layers on four qubits. The circuit has 8 layers and ends by measuring all qubits.

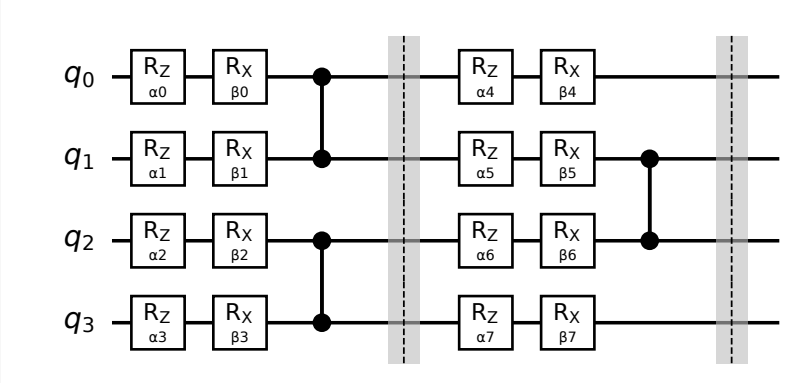


Figure 1.12: Two layers of a brick-wall circuit on four qubits.

(a) Count the native operations of each kind, and estimate the probability that a run of the circuit has no error at all. Which kind of operation contributes most to the failures?

Hint: Which gates in Figure 1.12 need a pulse?

(b) How does the answer to (a) change on a device without virtual Z rotations, where each R_Z needs a pulse with the same error as PRX?

(c) Roughly how many layers can the circuit have before the probability of a run without error drops below one half?

Hint: $(1-p)^n \approx e^{-np}$ for small p.

Exercise 1.9 (Readout errors bias the estimate)**.** Suppose that the readout of each qubit reports the wrong bit with probability p, independently in every shot and for both outcomes.

(a) Show that the recorded sign s of a single-qubit Z measurement has expectation value $(1-2p)\langle Z\rangle$.

(b) For a Pauli string P acting on k qubits, show that the recorded sign has expectation value $(1-2p)^k\langle P\rangle$.

Hint: The recorded sign is the true sign times one independent factor ± 1 per qubit.

(c) Take $p = 0.02$, $k = 4$ and $\langle P\rangle = 0.5$. Compare the bias of $\bar{s}$ with its standard error for $N = 10^3$ and for $N = 10^5$ shots. When do more shots stop helping?

Hint: The standard error is given by Equation 1.9.

(d) If p is known, how can the estimate be corrected? What does the correction do to the standard error?

Exercise 1.10 (Measuring in other bases)**.** A device has the native gates PRX and CZ, and measures in the computational basis.

(a) Which PRX pulses must be applied before the measurement to estimate $\langle X \otimes Y \otimes Z \rangle$?

Hint: On each qubit, you need a rotation U with $UP_jU^\dagger = Z$; see Exercise 1.3.

(b) Can $\langle X \otimes I \rangle$ and $\langle X \otimes X \rangle$ be estimated from the same shots? And $\langle X \otimes X \rangle$ and $\langle Z \otimes Z \rangle$, if only single-qubit rotations are allowed before the measurement?

1.7 Further reading

For superconducting qubits, Krantz et al. (2019) is a thorough guide written for engineers; for trapped ions, see Bruzewicz et al. (2019). Henriet et al. (2020), Slussarenko and Pryde (2019) and Burkard et al. (2023) review neutral atoms, photonics and spin qubits. Preskill (2018) introduced the NISQ era, and Nielsen and Chuang (2010, chap. 4) covers the circuit model.

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A Glossary

calibration Measuring and adjusting the control pulses of a device so that its gates do what they should. Providers publish calibration data: the error rates of each qubit and gate at the last calibration.

coherence time How long a qubit keeps its quantum state, in particular a superposition, before noise from its surroundings destroys it.

coherent error A systematic error in which a gate applies the wrong unitary, for example a rotation by a slightly wrong angle, in the same way every time.

coupler A circuit element between two superconducting qubits through which they interact. A tunable coupler can switch the interaction on and off.

coupling graph Graph whose vertices are qubits and whose edges are the pairs on which a native two-qubit gate is available.

cross-resonance A two-qubit gate mechanism for superconducting qubits with fixed couplings: one qubit is driven at the frequency of its neighbour.

crosstalk An unwanted effect of an operation on one qubit on other qubits, for example through couplings that are never fully switched off.

dephasing Loss of the relative phase of a superposition, for example because the qubit frequency fluctuates. Its characteristic time is called T2.

depth The number of layers of gates in a circuit that must run one after another. Gates on different qubits can share a layer.

digital quantum computer A quantum computer that runs circuits: sequences of gates from a fixed universal set, applied to qubits and followed by measurements. Also called gate-based, as opposed to analog devices that imitate the time evolution of a physical system directly.

dilution refrigerator A refrigerator that reaches about 10 millikelvin, cold enough that heat no longer excites superconducting qubits.

dispersive readout Readout of a superconducting qubit through a resonator whose frequency depends slightly on the qubit state. A weak microwave probe tells the two states apart.

DiVincenzo criteria A checklist for building a quantum computer: scalable, well-characterised qubits; preparation of a known state; long coherence; a universal set of gates; measurement of individual qubits.

error mitigation Methods that reduce the effect of noise on estimated quantities without correcting the errors, typically by running extra or modified circuits and combining their results. They cost extra shots.

estimator A rule that turns measured data into an estimate of a quantity, for example the sample mean of the outcomes as an estimate of an expectation value.

fidelity A number between 0 and 1 measuring how close an operation or state is to the intended one. Informally, one minus the probability of an error. Defined precisely in the chapter on noise.

ion An atom with one or more electrons removed, so that it carries an electric charge and can be held by electric fields.

ion trap A device that holds ions in vacuum with electric fields, typically lined up in a chain a few micrometres apart.

Josephson junction A thin insulating barrier between two superconductors. In a circuit it makes the energy levels unevenly spaced, which is what turns the circuit into a usable qubit.

laser cooling Slowing the motion of atoms or ions with laser light until they are almost at rest.

leakage An error in which a qubit leaves its two qubit levels, for example when a transmon is pushed into its third level.

logical qubit A qubit of information encoded in many physical qubits by an error-correcting code. Its error rate can be much lower than that of the physical qubits.

Mølmer-Sørensen gate The standard two-qubit gate of trapped ions. Laser beams couple two ions to the shared vibrations of the chain, producing an entangling rotation of any chosen angle.

native gate A gate the hardware implements directly with a calibrated control pulse. Every other gate must be compiled into native gates.

NISQ Noisy intermediate-scale quantum: devices with tens to thousands of qubits whose errors are not corrected, so noise limits the size of the circuits they can run.

optical tweezers Tightly focused laser beams that hold single neutral atoms in place and can move them.

Pauli string A tensor product of single-qubit Pauli matrices (I, X, Y or Z), one for each qubit, such as ZZ or XIY. Hermitian, and squares to the identity.

phonon A quantum of vibration. In an ion trap, phonons are the quanta of the vibrations that the whole chain of ions shares.

phonon bus The shared vibrations of a chain of trapped ions, used as a common channel through which any two ions can interact in a two-qubit gate.

physical qubit A qubit as implemented in hardware, for example one transmon or one ion, with all its imperfections.

pulse A short burst of microwave or laser radiation at a chosen frequency, switched on and off smoothly, used to drive a qubit. Its frequency selects which qubit responds; its area and phase set the rotation.

QCCD Quantum charge-coupled device: a trapped-ion architecture in which ions are moved between zones of the trap, so that any two ions can be brought together for a gate.

quantum dot A tiny region in a semiconductor that can trap single electrons.

quantum error correction Encoding quantum information redundantly in many physical qubits, so that errors can be detected and corrected without disturbing the encoded information.

relaxation Loss of energy by a qubit: the excited state decays to the ground state. Its characteristic time is called T1.

resonance Radiation drives a transition between two energy levels only when its frequency matches their energy difference divided by Planck's constant. This is how individual qubits are addressed.

Rydberg state A state of an atom in which one electron is excited very far from the nucleus. Atoms in Rydberg states interact strongly over micrometre distances.

shot One run of a circuit: prepare the qubits, apply the gates, measure. Returns one bit string.

spin An intrinsic two-level property of an electron (up or down), which can serve as a qubit.

SWAP A two-qubit gate that exchanges the states of two qubits. It costs three CNOTs.

threshold The physical error rate per operation below which error correction helps: below it, larger codes give lower logical error rates.

transmon The most common superconducting qubit: a small circuit with a Josephson junction, whose two lowest energy levels form the qubit.

transpilation Compilation of a circuit into the native gates and coupling graph of a device, including inserting SWAPs and simplifying.

virtual Z gate A Z rotation implemented by shifting the phase of all later drive pulses instead of applying a pulse. It takes no time and adds no error.